

维修手册_TC820LTYsi_故障诊断数据

维修手册知识库拆分数据 · 全字段 PDF 导出

源数据文件	维修手册_TC820LTYsi_故障诊断数据.jsonl
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文档来源	Southwestern Industries · TRAK TCsi with SINUMERIK ONE 维修手册
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记录类别分布

记录类别	条数	占比
故障/报警 (fault)	16	100.0%

知识类型分布

knowledge_type	条数
diagnosis	14
diagnostic_procedure	2

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#1 lty-source-tcsi-diagnostic-mcp-dark			故障/报警
部件	machine control panel	章节	Control Troubleshooting
起始页	72	安全等级	normal
知识类型	diagnosis	适用机型	TC820LTYsi

摘要 | MCP 闪烁或黑屏时，先检查 PPU 到 MCP 的网络电缆连接。

A • 标识与来源

记录ID	lty-source-tcsi-diagnostic-mcp-dark
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	4.2.1
章节	Control Troubleshooting
起始页	72
结束页	72

C • 分类与治理

记录类别	fault
知识类型	diagnosis
部件	machine control panel
安全等级	normal
可入库	是
需人工复核	否

D • 业务明细

故障来源	manual_diagnostic_section
故障现象	MCP 闪烁、黑屏或无响应；可能显示 “Failure of machine control panel 1” 。
原因	MCP 与 PPU 之间网络连接异常。
检查项	检查 PPU 到 MCP 的网络电缆连接
处置措施	确认网络电缆连接状态后再进行进一步维修。

适用机型	TC820LTYsi
E • 检索与原文	
中文摘要	MCP 闪烁或黑屏时，先检查 PPU 到 MCP 的网络电缆连接。
关键词	· MCP黑屏 · 控制面板 · 网络电缆 · PPU
检索文本	MCP 闪烁或黑屏时，先检查 PPU 到 MCP 的网络电缆连接。； MCP 闪烁、黑屏或无响应；可能显示 “Failure of machine control panel 1” 。； MCP 与 PPU 之间网络连接异常。； 确认网络电缆连接状态后再进行进一步维修。； Reported problem: a known-good TCsi machine starts to flash or has a dark, non-responsive MCP, possibly showing “Failure of machine control panel 1” . Check the network cable connection from PPU to MCP.； 检查 PPU 到 MCP 的网络电缆连接； MCP黑屏； 控制面板； 网络电缆； PPU
原文摘录	It is suggested to remove screen prints from the memory of the controller by highlighting one or more files and pressing “Delete” . There is limited memory space which should be left for programs. Figure 3.1e Edit Alarm texts 4.2 Control Troubleshooting 4.2.1 Flashing or Dark Machine Control Panel (MCP) Reported problem is from a known good TCsi machine that starts to flash or has a dark non responsive MCP. 1. Possible message shown as MCP fails or flashes “Failure of machine control panel 1” . 2. Check network cable connection from PPU to MCP see figure 3.2.1. Figure 3.2.1 Pendant Rear View

#2 lty-source-tcsi-diagnostic-handwheel			故障/报警
部件	manual pulse generator	章节	Control Troubleshooting
起始页	73	安全等级	normal
知识类型	diagnosis	适用机型	TC820LTYsi

摘要 | 手轮不工作时依次检查手轮本体、后盖线束和 X143 连接器线路连续性。

A • 标识与来源

记录ID	lty-source-tcsi-diagnostic-handwheel
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	4.2.2

章节	Control Troubleshooting
起始页	73
结束页	73

C • 分类与治理

记录类别	fault
知识类型	diagnosis
部件	manual pulse generator
安全等级	normal
可入库	是
需人工复核	否

D • 业务明细

故障来源	manual_diagnostic_section
故障现象	EHW/手轮原本正常但停止工作。
原因	EHW 本体、线束或 X143 连接器线路异常。
检查项	· 检查 EHW remote 状态 · 拆下面板后盖检查 EHW 到 pendant 的线束 · 检查 MPG 到 pendant 底部及 X143 连接器线路是否连续、无损坏且紧固
处置措施	按电源安全要求检查并恢复连接；若线路正常仍无效，升级维修。
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	手轮不工作时依次检查手轮本体、后盖线束和 X143 连接器线路连续性。
关键词	· 手轮 · EHW · MPG · X143 · 不工作
检索文本	手轮不工作时依次检查手轮本体、后盖线束和 X143 连接器线路连续性。； EHW/手轮原本正常但停止工作。； EHW 本体、线束或 X143 连接器线路异常。； 按电源安全要求检查并恢复连接；若线路正常仍无效，升级维修。； For an EHW that has stopped working, check the EHW remote, inspect wires from the EHW to the pendant, and verify the wires from the MPG to the individual wires on the X143 connector are continuous, undamaged and tight.； 检查 EHW remote 状态；拆下面板后盖检查 EHW 到 pendant 的线束；检查 MPG 到 pendant 底部及 X143 连接器线路是否连续、无损坏且紧固；手轮； EHW； MPG； X143； 不工作
原文摘录	4.2.2 Handwheels Do Not Work Reported problem from a known good TCsi machine with working EHW that stop working. 1. Check condition of the EHW remote. 2. Remove rear cover of pendant and check wires from the EHW to pendant. 3. Check wires from the EHW to pendant. If needed open pendant rear cover and check wires from bottom of pendant to the individual wires on X143 connector as shown in figure 3.2.2. Verify wires are continuous, undamaged and tight from the MPG to the pendant. Figure 3.2.2 Manual Pulse Generator wiring 4.2.3 Check Defective Drive System Using System Topology Using system Topology Screens will help detect drive system problems. SINUMERIK system screens have the capability to show in real time the system topology. From the left VSK, press “Setup” , then from HSK press “Drive System” and “Topology” which will display the actual topology layout of the system in real time see figure 3.2.3a. Figure 3.2.3a Topology Display

#3 lty-source-tcsi-diagnostic-topology			故障/报警
部件	drive system	章节	Control Troubleshooting
起始页	73	安全等级	normal
知识类型	diagnostic_procedure	适用机型	TC820LTYsi

摘要 | 通过 SINUMERIK 的 Drive System/Topology/Topology comparison 查看实时驱动拓扑，异常组件会变色或变灰。

A • 标识与来源

记录ID	lty-source-tcsi-diagnostic-topology
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	4.2.3
章节	Control Troubleshooting
起始页	73
结束页	73

C • 分类与治理

记录类别	fault
知识类型	diagnostic_procedure
部件	drive system
安全等级	normal
可入库	是
需人工复核	否

D • 业务明细

故障来源	manual_diagnostic_section
故障现象	需要定位驱动系统故障。
原因	驱动拓扑中的异常组件会在实时拓扑对比中变色或变灰。

检查项	· 左侧 VSK 选择 Setup · HSK 选择 Drive System → Topology · 选择 Topology comparison，观察实际拓扑与系统设计差异
处置措施	根据变色或变灰组件缩小故障范围，并结合报警和 I/O 证据继续检查。
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	通过 SINUMERIK 的 Drive System/Topology/Topology comparison 查看实时驱动拓扑，异常组件会变色或变灰。
关键词	· Topology · 驱动系统 · 变灰 · Drive System · 诊断
检索文本	通过 SINUMERIK 的 Drive System/Topology/Topology comparison 查看实时驱动拓扑，异常组件会变色或变灰。；需要定位驱动系统故障。；驱动拓扑中的异常组件会在实时拓扑对比中变色或变灰。；根据变色或变灰组件缩小故障范围，并结合报警和 I/O 证据继续检查。；Use Setup → Drive System → Topology, then Topology comparison. When the system is working correctly the actual topology matches the design; problem components are shown by changed colors or as grayed-out components.; 左侧 VSK 选择 Setup; HSK 选择 Drive System → Topology; 选择 Topology comparison，观察实际拓扑与系统设计差异; Topology; 驱动系统; 变灰; Drive System; 诊断
原文摘录	4.2.2 Handwheels Do Not Work Reported problem from a known good TCsi machine with working EHW that stop working. 1. Check condition of the EHW remote. 2. Remove rear cover of pendant and check wires from the EHW to pendant. 3. Check wires from the EHW to pendant. If needed open pendant rear cover and check wires from bottom of pendant to the individual wires on X143 connector as shown in figure 3.2.2. Verify wires are continuous, undamaged and tight from the MPG to the pendant. Figure 3.2.2 Manual Pulse Generator wiring 4.2.3 Check Defective Drive System Using System Topology Using system Topology Screens will help detect drive system problems. SINUMERIK system screens have the capability to show in real time the system topology. From the left VSK, press “Setup” , then from HSK press “Drive System” and “Topology” which will display the actual topology layout of the system in real time see figure 3.2.3a. Figure 3.2.3a Topology Display

#4 lty-source-tcsi-diagnostic-io			故障/报警
部件	machine I/O	章节	TRAK Diagnostic Screens / I/O Diagnostic
起始页	77	安全等级	normal
知识类型	diagnostic_procedure	适用机型	TC820LTYsi

摘要 | I/O Diagnostic 可显示机器当前输入输出状态，用于定位开关、反馈和执行器问题。

A • 标识与来源

记录ID	lty-source-tcsi-diagnostic-io
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725

语言	en
产品系列	TRAK TCsi
B • 型号与位置	
设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	4.5.2
章节	TRAK Diagnostic Screens / I/O Diagnostic
起始页	77
结束页	77

C • 分类与治理

记录类别	fault
知识类型	diagnostic_procedure
部件	machine I/O
安全等级	normal
可入库	是
需人工复核	否

D • 业务明细

故障来源	manual_diagnostic_section
故障现象	某一机械功能动作异常，需要判断输入输出状态。
原因	现场输入、输出或反馈状态与预期不一致。
检查项	· Diag → TRAK Diag. → Input/Output · 观察当前输入输出状态 · 在确认安全的前提下监控相关 I/O 变化
处置措施	把 I/O 诊断结果提供给 TRAK Customer Service，结合对应 PLC 报警、接线图和组件检查。
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	I/O Diagnostic 可显示机器当前输入输出状态，用于定位开关、反馈和执行器问题。
关键词	· I/O诊断 · 输入输出 · 反馈 · TRAK Diag
检索文本	I/O Diagnostic 可显示机器当前输入输出状态，用于定位开关、反馈和执行器问题。； 某一机械功能动作异常，需要判断输入输出状态。； 现场输入、输出或反馈状态与预期不一致。； 把 I/O 诊断结果提供给 TRAK Customer Service，结合对应 PLC 报警、接线图和组件检查。； The I/O Diagnostic screen shows the current state of machine inputs and outputs and can be used to troubleshoot different functions and determine where an issue may be. TRAK customer service may ask for this screen during troubleshooting.; Diag → TRAK Diag. → Input/Output; 观察当前输入输出状态; 在确认安全的前提下监控相关 I/O 变化; I/O诊断; 输入输出; 反馈; TRAK Diag

原文摘录

4.5 TRAK Diagnostic Screens 4.5.1 Run Times The Run Time screen is found by pressing the left VSK “Diag” button, from “Diag” page 1 press “TRAK Diag.” button on the right side of the HSK. The operator may scroll down to find various data listed below see figure 3.5.1a. Figure 3.5.1a TRAK Diag Control will display the Run Times screen for analysis see figure 3.5.1b. Figure 3.5.1b Run Times 4.5.2 I/O Diagnostic The I/O Diagnostic screen is found by pressing the left VSK “Diag” button, from “Diag” page 1 press “TRAK Diag.” button on the right side of the HSK. From the “Run Times” press “Input/Output.” on the left side of the HSK see figure 3.5.2a. Figure 3.5.2a I/O Diagnostic

#5 lty-source-tcsi-diagnostic-taper			故障/报警
部件	machine setup and alignment	章节	Problems Relating to Machining Results
起始页	79	安全等级	normal
知识类型	diagnosis	适用机型	TC820LTYsi

摘要 | 直径在 6 in 长度上差异超过 0.0008 in 可视为锥度，优先检查调平、夹持和对中。

A • 标识与来源

记录ID	lty-source-tcsi-diagnostic-taper
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	4.6
章节	Problems Relating to Machining Results
起始页	79
结束页	79

C • 分类与治理

记录类别	fault
知识类型	diagnosis
部件	machine setup and alignment
安全等级	normal
可入库	是
需人工复核	否

D • 业务明细

故障来源	manual_diagnostic_section
故障现象	工件出现锥度。
原因	机床未正确调平、机床失准或工件/夹持设置问题。
检查项	· 用 0.0001 in 千分尺测量 · 直径差超过 0.0008 in over 6 in 时视为切削锥度 · 检查调平、夹持和机床对中
处置措施	按调平、夹持和对中章节检查；碰撞后失准应联系 TRAK。
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	直径在 6 in 长度上差异超过 0.0008 in 可视为锥度，优先检查调平、夹持和对中。
关键词	· 锥度 · taper · 调平 · 对中 · 0.0008
检索文本	直径在 6 in 长度上差异超过 0.0008 in 可视为锥度，优先检查调平、夹持和对中。；工件出现锥度。；机床未正确调平、机床失准或工件/夹持设置问题。；按调平、夹持和对中章节检查；碰撞后失准应联系 TRAK。；Parts are considered to be cutting on a taper if there is a difference in diameter of more than 0.0008 in over 6 in. Measure with a 0.0001 in micrometer. Check machine leveling, workholding and alignment.；用 0.0001 in 千分尺测量；直径差超过 0.0008 in over 6 in 时视为切削锥度；检查调平、夹持和机床对中；锥度；taper；调平；对中；0.0008
原文摘录	the collision detection message call TRAK Customer Service at (800)-367-3165 to check over the machine before clearing the message. Figure 3.5.3b Collision Detection 4.6 Problems Relating to Machining Results Problems with machining results can be caused by a number of variables including: speeds, feeds, tooling, machine setup, alignment, programing, and chatter. • Parts are considered to be cutting on a taper if there is a difference in diameter of more than 0.0008” over 6” . This is best measured by using a .0001” micrometer. • Diameters are considered out of round if they are not within .0002” TIR. • Threaded parts may have no thread, be cross-threaded, or have the incorrect thread after completion of the threading event. Possible Cause Check This Machine set-up problem Machine not leveled properly/feet are not equally supporting weight. See Leveling Chapter 2. Machine set-up problem Workpiece moves in the chuck. See Machine Tool Machine Tool and Setup problem – poor surface & Setup. See Chuck Chapter 2. finish Check the type of material being cut, type and size of cutting tool, tool on centerline, RPM, and feed Machine Tool and Setup problem – Excessive rate, depth of cut. Also check the condition of the chatter cutter to verify that the cutter is not dull. Use coolant. Tool bit improperly ground or not on center, avoid extreme negative rake inserts, tool overhang too great, tool deflection, improper feeds and speeds, X gib loose, work improperly supported, machine out of level

#6 lty-source-tcsi-diagnostic-roundness			故障/报警
部件	machining result	章节	Problems Relating to Machining Results
起始页	79	安全等级	normal
知识类型	diagnosis	适用机型	TC820LTYsi

摘要 | 直径不在 0.0002 in TIR 内时视为不圆，检查夹持、刀具、主轴及对中。

A • 标识与来源

记录ID	lty-source-tcsi-diagnostic-roundness
文档ID	trak-tcsi-service-34242-v022725

文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
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语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	4.6
章节	Problems Relating to Machining Results
起始页	79
结束页	79

C • 分类与治理

记录类别	fault
知识类型	diagnosis
部件	machining result
安全等级	normal
可入库	是
需人工复核	否

D • 业务明细

故障来源	manual_diagnostic_section
故障现象	工件直径不圆。
原因	夹持、刀具、主轴、对中或机床状态异常。
检查项	· 测量 TIR · 直径不在 0.0002 in TIR 内时视为不圆 · 检查夹持、刀具、主轴和机床对中
处置措施	按加工结果、刀具和机床对中项目逐项排查。
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	直径不在 0.0002 in TIR 内时视为不圆，检查夹持、刀具、主轴及对中。
关键词	· 不圆 · out of round · TIR · 0.0002

检索文本	直径不在 0.0002 in TIR 内时视为不圆，检查夹持、刀具、主轴及对中。；工件直径不圆。；夹持、刀具、主轴、对中或机床状态异常。；按加工结果、刀具和机床对中项目逐项排查。； Diameters are considered out of round if they are not within 0.0002 in TIR. Check workholding, tooling, spindle condition and machine alignment.；测量 TIR；直径不在 0.0002 in TIR 内时视为不圆；检查夹持、刀具、主轴和机床对中；不圆；out of round；TIR；0.0002
原文摘录	the collision detection message call TRAK Customer Service at (800)-367-3165 to check over the machine before clearing the message. Figure 3.5.3b Collision Detection 4.6 Problems Relating to Machining Results Problems with machining results can be caused by a number of variables including: speeds, feeds, tooling, machine setup, alignment, programing, and chatter. • Parts are considered to be cutting on a taper if there is a difference in diameter of more than 0.0008” over 6” . This is best measured by using a .0001” micrometer. • Diameters are considered out of round if they are not within .0002” TIR. • Threaded parts may have no thread, be cross-threaded, or have the incorrect thread after completion of the threading event. Possible Cause Check This Machine set-up problem Machine not leveled properly/feet are not equally supporting weight. See Leveling Chapter 2. Machine set-up problem Workpiece moves in the chuck. See Machine Tool Machine Tool and Setup problem – poor surface & Setup. See Chuck Chapter 2. finish Check the type of material being cut, type and size of cutting tool, tool on centerline, RPM, and feed Machine Tool and Setup problem – Excessive rate, depth of cut. Also check the condition of the chatter cutter to verify that the cutter is not dull. Use coolant. Tool bit improperly ground or not on center, avoid extreme negative rake inserts, tool overhang too great, tool deflection, improper feeds and speeds, X gib loose, work improperly supported, machine out of level

#7 lty-source-tcsi-diagnostic-surface-finish			故障/报警
部件	cutting setup	章节	Problems Relating to Machining Results
起始页	79	安全等级	normal
知识类型	diagnosis	适用机型	TC820LTYsi

摘要 | 表面质量差优先检查材料、刀具中心高、RPM、进给、切深、刀具磨损和冷却液。

A • 标识与来源

记录ID	lty-source-tcsi-diagnostic-surface-finish
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	4.6
章节	Problems Relating to Machining Results

起始页	79
结束页	79

C • 分类与治理

记录类别	fault
知识类型	diagnosis
部件	cutting setup
安全等级	normal
可入库	是
需人工复核	否

D • 业务明细

故障来源	manual_diagnostic_section
故障现象	表面粗糙度差。
原因	材料、刀具类型/尺寸、刀尖中心高、转速、进给、切深或冷却液设置不合适。
检查项	· 检查材料和刀具类型尺寸 · 确认刀具在中心线上且无钝化 · 检查 RPM、进给率、切深和冷却液
处置措施	调整加工参数和刀具状态；排查工件支撑和机床调平。
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	表面质量差优先检查材料、刀具中心高、RPM、进给、切深、刀具磨损和冷却液。
关键词	· 表面粗糙 · surface finish · RPM · 进给 · 刀具磨损
检索文本	表面质量差优先检查材料、刀具中心高、RPM、进给、切深、刀具磨损和冷却液。；表面粗糙度差。；材料、刀具类型/尺寸、刀尖中心高、转速、进给、切深或冷却液设置不合适。；调整加工参数和刀具状态；排查工件支撑和机床调平。； For poor surface finish, check the material, type and size of cutting tool, tool on centerline, RPM, feed rate, depth of cut, cutter condition and coolant. Also check work support and machine level.；检查材料和刀具类型尺寸；确认刀具在中心线上且无钝化；检查 RPM、进给率、切深和冷却液；表面粗糙； surface finish； RPM；进给；刀具磨损
原文摘录	the collision detection message call TRAK Customer Service at (800)-367-3165 to check over the machine before clearing the message. Figure 3.5.3b Collision Detection 4.6 Problems Relating to Machining Results Problems with machining results can be caused by a number of variables including: speeds, feeds, tooling, machine setup, alignment, programing, and chatter. • Parts are considered to be cutting on a taper if there is a difference in diameter of more than 0.0008” over 6” . This is best measured by using a .0001” micrometer. • Diameters are considered out of round if they are not within .0002” TIR. • Threaded parts may have no thread, be cross-threaded, or have the incorrect thread after completion of the threading event. Possible Cause Check This Machine set-up problem Machine not leveled properly/feet are not equally supporting weight. See Leveling Chapter 2. Machine set-up problem Workpiece moves in the chuck. See Machine Tool Machine Tool and Setup problem – poor surface & Setup. See Chuck Chapter 2. finish Check the type of material being cut, type and size of cutting tool, tool on centerline, RPM, and feed Machine Tool and Setup problem – Excessive rate, depth of cut. Also check the condition of the chatter cutter to verify that the cutter is not dull. Use coolant. Tool bit improperly ground or not on center, avoid extreme negative rake inserts, tool overhang too great, tool deflection, improper feeds and speeds, X gib loose, work improperly supported, machine out of level

#8 lty-source-tcsi-diagnostic-chatter			故障/报警
部件	cutting setup	章节	Problems Relating to Machining Results
起始页	79	安全等级	normal
知识类型	diagnosis	适用机型	TC820LTYsi

摘要 | 振刀排查刀具几何与磨损、悬伸/偏转、进给转速、镶条、支撑和调平。

A • 标识与来源

记录ID	lty-source-tcsi-diagnostic-chatter
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	4.6
章节	Problems Relating to Machining Results
起始页	79
结束页	79

C • 分类与治理

记录类别	fault
知识类型	diagnosis
部件	cutting setup
安全等级	normal
可入库	是
需人工复核	否

D • 业务明细

故障来源	manual_diagnostic_section
故障现象	加工中振刀或颤振过大。
原因	刀具磨损/研磨不当、刀具悬伸过大、刀具偏转、进给转速不当、X轴镶条松、工件支撑不足或机床未调平。

检查项	· 检查刀尖研磨和中心高 · 检查负前角刀片、刀具悬伸和偏转 · 检查进给/转速、X 轴镶条、工件支撑和调平
处置措施	调整刀具和加工参数，保证支撑和调平；必要时检查镶条。
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	振刀排查刀具几何与磨损、悬伸/偏转、进给转速、镶条、支撑和调平。
关键词	· 振刀 · 颤振 · chatter · 镶条 · 刀具悬伸
检索文本	振刀排查刀具几何与磨损、悬伸/偏转、进给转速、镶条、支撑和调平。；加工中振刀或颤振过大。；刀具磨损/研磨不当、刀具悬伸过大、刀具偏转、进给转速不当、X 轴镶条松、工件支撑不足或机床未调平。；调整刀具和加工参数，保证支撑和调平；必要时检查镶条。；Excessive chatter can result from an improperly ground or off-center tool, extreme negative-rake inserts, excessive tool overhang, tool deflection, improper feeds and speeds, a loose X gib, poor work support or a machine out of level.；检查刀尖研磨和中心高；检查负前角刀片、刀具悬伸和偏转；检查进给/转速、X 轴镶条、工件支撑和调平；振刀；颤振；chatter；镶条；刀具悬伸
原文摘录	the collision detection message call TRAK Customer Service at (800)-367-3165 to check over the machine before clearing the message. Figure 3.5.3b Collision Detection 4.6 Problems Relating to Machining Results Problems with machining results can be caused by a number of variables including: speeds, feeds, tooling, machine setup, alignment, programing, and chatter. • Parts are considered to be cutting on a taper if there is a difference in diameter of more than 0.0008” over 6” . This is best measured by using a .0001” micrometer. • Diameters are considered out of round if they are not within .0002” TIR. • Threaded parts may have no thread, be cross-threaded, or have the incorrect thread after completion of the threading event. Possible Cause Check This Machine set-up problem Machine not leveled properly/feet are not equally supporting weight. See Leveling Chapter 2. Machine set-up problem Workpiece moves in the chuck. See Machine Tool Machine Tool and Setup problem – poor surface & Setup. See Chuck Chapter 2. finish Check the type of material being cut, type and size of cutting tool, tool on centerline, RPM, and feed Machine Tool and Setup problem – Excessive rate, depth of cut. Also check the condition of the chatter cutter to verify that the cutter is not dull. Use coolant. Tool bit improperly ground or not on center, avoid extreme negative rake inserts, tool overhang too great, tool deflection, improper feeds and speeds, X gib loose, work improperly supported, machine out of level

#9 lty-source-tcsi-diagnostic-thermal-growth			故障/报警
部件	machine thermal condition	章节	Problems Relating to Machining Results
起始页	80	安全等级	normal
知识类型	diagnosis	适用机型	TC820LTYsi

摘要 | 尺寸热漂移时先预热机床，再检查刀具磨损、偏置、坐标系、支撑和程序绝对/增量值。

A • 标识与来源

记录ID	lty-source-tcsi-diagnostic-thermal-growth
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf

制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	4.6
章节	Problems Relating to Machining Results
起始页	80
结束页	80

C • 分类与治理

记录类别	fault
知识类型	diagnosis
部件	machine thermal condition
安全等级	normal
可入库	是
需人工复核	否

D • 业务明细

故障来源	manual_diagnostic_section
故障现象	加工尺寸随运行时间变化或尺寸不稳定。
原因	热增长、未预热、夹持/支撑不当、刀具磨损或程序/偏置问题。
检查项	· 预热机床使部件达到正常工作温度 · 检查刀具、偏置、坐标系和工件支撑 · 检查活动顶尖磨损和悬伸
处置措施	按 Warm-Up Procedure 运行机床，并复核程序尺寸及绝对/增量值。
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	尺寸热漂移时先预热机床，再检查刀具磨损、偏置、坐标系、支撑和程序绝对/增量值。
关键词	· 热增长 · thermal growth · 预热 · 尺寸漂移 · 偏置

检索文本	尺寸热漂移时先预热机床，再检查刀具磨损、偏置、坐标系、支撑和程序绝对/增量值。；加工尺寸随运行时间变化或尺寸不稳定。；热增长、未预热、夹持/支撑不当、刀具磨损或程序/偏置问题。；按 Warm-Up Procedure 运行机床，并复核程序尺寸及绝对/增量值。；Thermal growth may cause machining-result problems. Warm up the machine and bring components to normal operating temperature. Check tooling, wear, offsets, coordinate system, workpiece support, live center condition and overhang; verify programmed dimensions and absolute/incremental values.；预热机床使部件达到正常工作温度；检查刀具、偏置、坐标系和工件支撑；检查活动顶尖磨损和悬伸；热增长；thermal growth；预热；尺寸漂移；偏置
原文摘录	Possible Cause Check This Tool Probe Set up Tooling problem Make sure the tool probe is set up correctly and calibrated accurately. See Tool Probe section 6.3. Programming Error Improper tooling, loose tooling, tool wear, offsets, Thermal Growth coordinate system, workpiece not properly supported, worn or damaged live center, overhang Gibs too tight or loose from chuck headstock or tailstock. incorrect speeds Inadequate or no Lubrication to Ballscrews and and Feeds. Way surfaces Programmed dimensions are not correct. Check absolute and incremental values. Way surfaces are pocked, scarred, or excessively You must warm the machine up and bring all the worn components to normal operation temperature. See Machine Alignment Warm-Up section 2.17. See Gib Adjustment Chapter 4. Binding or looseness in the drive train or bearings Make sure all the Way surfaces are getting proper lubrication. Check to make sure that the lube pump is functioning properly. Also check for any pinched or blocked oil lines. See Lubrication Chapter 2. Visually check the condition of all the Way surfaces. For machines that may have excessively worn Way surfaces check lubrication to affected areas. After a crash the machine may become out of alignment. See section 4.4. Call TRAK Customer Service at (800)-367-3165 Call TRAK Customer Service at (800)-367-3165 4.6.1 Problems Relating to Live Tooling Machining Results Problems with live tooling machining results can be; machine setup, alignment, and component failure. • Live tooling assembly may have excessive noise or vibrations. • Live tooling assembly is getting too hot. • Live tool may be faulty. Possible Cause Check This The drive is out of alignment. A drive bearing has failed. Check the alignment of the live tooling drive. See section 5.2.2.5. The drive belt is faulty Perform vibration test on the live tool and check for excessive vibration. Call TRAK Customer Service at (800)-367-3165 Check live tooling spindle is turning and inspect the live tooling drive belt. 4.7 Problems Relating to the Machine Tool and Motion of the Machine Problems with the motion of the machine will mainly be caused by: machine setup, alignment, and component failure. • Axis may slow down or move at a feedrate lower than rapid or than the programmed feedrate. • While under motor power, the motion may appear to be "rough" or jerky" . • While under motor power, there may be vibration or noise coming from the X or Z-axis.

#10 lty-source-tcsi-diagnostic-lube-motion			故障/报警
部件	ways and ballscrews	章节	Problems Relating to Machine Tool and Motion
起始页	80	安全等级	normal
知识类型	diagnosis	适用机型	TC820LTysi

摘要 | 轴慢、卡滞、振动或重复性差时优先查润滑/油路、镶条、反向间隙、对中及传动绑定。

A • 标识与来源

记录ID	lty-source-tcsi-diagnostic-lube-motion
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725

语言	en
产品系列	TRAK TCsi
B • 型号与位置	
设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	4.7
章节	Problems Relating to Machine Tool and Motion
起始页	80
结束页	80

C • 分类与治理

记录类别	fault
知识类型	diagnosis
部件	ways and ballscrews
安全等级	normal
可入库	是
需人工复核	否

D • 业务明细

故障来源	manual_diagnostic_section
故障现象	轴运动慢、运动不平顺、噪声/振动或定位重复性差。
原因	润滑不足、镶条过紧/过松、传动绑定、反向间隙、机床失准或电缆/驱动故障。
检查项	· 检查所有滑轨润滑和润滑泵 · 检查油管是否夹扁或堵塞 · 检查镶条、反向间隙和重复性/定位精度 · 碰撞后检查机床对中
处置措施	恢复充分润滑，调整镶条并复核反向间隙；驱动、轴承或对中问题联系 TRAK。
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	轴慢、卡滞、振动或重复性差时优先查润滑/油路、镶条、反向间隙、对中及传动绑定。
关键词	· 轴运动 · 定位精度 · 重复性 · 润滑 · 反向间隙 · 绑定

检索文本	轴慢、卡滞、振动或重复性差时优先查润滑/油路、镶条、反向间隙、对中及传动绑定。；轴运动慢、运动不平顺、噪声/振动或定位重复性差。；润滑不足、镶条过紧/过松、传动绑定、反向间隙、机床失准或电缆/驱动故障。；恢复充分润滑，调整镶条并复核反向间隙；驱动、轴承或对中问题联系 TRAK。；Motion problems may be caused by inadequate lubrication, gibs too tight or loose, binding or looseness in the drive train, calibration/backlash, machine alignment, defective cables, spindle bearing noise, or drive/motor/encoder failure. Check way lubrication, pump and oil lines; reset backlash and check repeatability and positional accuracy.; 检查所有滑轨润滑和润滑泵；检查油管是否夹扁或堵塞；检查镶条、反向间隙和重复性/定位精度；碰撞后检查机床对中；轴运动；定位精度；重复性；润滑；反向间隙；绑定
原文摘录	Possible Cause Check This Tool Probe Set up Tooling problem Make sure the tool probe is set up correctly and calibrated accurately. See Tool Probe section 6.3. Programming Error Improper tooling, loose tooling, tool wear, offsets, Thermal Growth coordinate system, workpiece not properly supported, worn or damaged live center, overhang Gibs too tight or loose from chuck headstock or tailstock. incorrect speeds Inadequate or no Lubrication to Ballscrews and and Feeds. Way surfaces Programmed dimensions are not correct. Check absolute and incremental values. Way surfaces are pocked, scarred, or excessively You must warm the machine up and bring all the worn components to normal operation temperature. See Machine Alignment Warm-Up section 2.17. See Gib Adjustment Chapter 4. Binding or looseness in the drive train or bearings Make sure all the Way surfaces are getting proper lubrication. Check to make sure that the lube pump is functioning properly. Also check for any pinched or blocked oil lines. See Lubrication Chapter 2. Visually check the condition of all the Way surfaces. For machines that may have excessively worn Way surfaces check lubrication to affected areas. After a crash the machine may become out of alignment. See section 4.4. Call TRAK Customer Service at (800)-367-3165 Call TRAK Customer Service at (800)-367-3165 4.6.1 Problems Relating to Live Tooling Machining Results Problems with live tooling machining results can be; machine setup, alignment, and component failure. • Live tooling assembly may have excessive noise or vibrations. • Live tooling assembly is getting too hot. • Live tool may be faulty. Possible Cause Check This The drive is out of alignment. A drive bearing has failed. Check the alignment of the live tooling drive. See section 5.2.2.5. The drive belt is faulty Perform vibration test on the live tool and check for excessive vibration. Call TRAK Customer Service at (800)-367-3165 Check live tooling spindle is turning and inspect the live tooling drive belt. 4.7 Problems Relating to the Machine Tool and Motion of the Machine Problems with the motion of the machine will mainly be caused by: machine setup, alignment, and component failure. • Axis may slow down or move at a feedrate lower than rapid or than the programmed feedrate. • While under motor power, the motion may appear to be "rough" or jerky" . • While under motor power, there may be vibration or noise coming from the X or Z-axis.

#11 lty-source-tcsi-diagnostic-spindle			故障/报警
部件	spindle	章节	Problems Relating to Machine Tool and Motion
起始页	80	安全等级	normal
知识类型	diagnosis	适用机型	TC820LTYsi

摘要 | 主轴无力、嗡鸣或不启动，检查皮带张力/共振、电缆、轴承和驱动电机编码器。

A • 标识与来源

记录ID	lty-source-tcsi-diagnostic-spindle
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725

语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	4.7
章节	Problems Relating to Machine Tool and Motion
起始页	80
结束页	80

C • 分类与治理

记录类别	fault
知识类型	diagnosis
部件	spindle
安全等级	normal
可入库	是
需人工复核	否

D • 业务明细

故障来源	manual_diagnostic_section
故障现象	主轴失去动力、持续嗡鸣或无法启动。
原因	皮带打滑/共振、轴承噪声、驱动/电机/编码器故障或电缆连接异常。
检查项	· 检查传动皮带对中、状态和张力 · 改变主轴速度观察振动和噪声是否降低 · 检查所有电缆连接 · 检查主轴轴承和驱动/电机/编码器
处置措施	先排查皮带、共振和连接；疑似轴承、驱动、电机或编码器故障时联系 TRAK。
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	主轴无力、嗡鸣或不启动，检查皮带张力/共振、电缆、轴承和驱动电机编码器。
关键词	· 主轴 · 嗡鸣 · 皮带 · 共振 · 编码器 · 不启动

检索文本	主轴无力、嗡鸣或不启动，检查皮带张力/共振、电缆、轴承和驱动电机编码器。；主轴失去动力、持续嗡鸣或无法启动。；皮带打滑/共振、轴承噪声、驱动/电机/编码器故障或电缆连接异常。；先排查皮带、共振和连接；疑似轴承、驱动、电机或编码器故障时联系 TRAK。；If the spindle loses power, hums constantly or does not turn on, check drive-belt alignment, condition and tension; speed up or slow down the spindle to reduce resonance; check cable connections; and investigate spindle bearing, drive, motor or encoder failure.；检查传动皮带对中、状态和张紧力；改变主轴速度观察振动和噪声是否降低；检查所有电缆连接；检查主轴轴承和驱动/电机/编码器；主轴；嗡鸣；皮带；共振；编码器；不启动
原文摘录	Possible Cause Check This Tool Probe Set up Tooling problem Make sure the tool probe is set up correctly and calibrated accurately. See Tool Probe section 6.3. Programming Error Improper tooling, loose tooling, tool wear, offsets, Thermal Growth coordinate system, workpiece not properly supported, worn or damaged live center, overhang Gibs too tight or loose from chuck headstock or tailstock. incorrect speeds Inadequate or no Lubrication to Ballscrews and and Feeds. Way surfaces Programmed dimensions are not correct. Check absolute and incremental values. Way surfaces are pocked, scarred, or excessively You must warm the machine up and bring all the worn components to normal operation temperature. See Machine Alignment Warm-Up section 2.17. See Gib Adjustment Chapter 4. Binding or looseness in the drive train or bearings Make sure all the Way surfaces are getting proper lubrication. Check to make sure that the lube pump is functioning properly. Also check for any pinched or blocked oil lines. See Lubrication Chapter 2. Visually check the condition of all the Way surfaces. For machines that may have excessively worn Way surfaces check lubrication to affected areas. After a crash the machine may become out of alignment. See section 4.4. Call TRAK Customer Service at (800)-367-3165 Call TRAK Customer Service at (800)-367-3165 4.6.1 Problems Relating to Live Tooling Machining Results Problems with live tooling machining results can be; machine setup, alignment, and component failure. • Live tooling assembly may have excessive noise or vibrations. • Live tooling assembly is getting too hot. • Live tool may be faulty. Possible Cause Check This The drive is out of alignment. A drive bearing has failed. Check the alignment of the live tooling drive. See section 5.2.2.5. The drive belt is faulty Perform vibration test on the live tool and check for excessive vibration. Call TRAK Customer Service at (800)-367-3165 Check live tooling spindle is turning and inspect the live tooling drive belt. 4.7 Problems Relating to the Machine Tool and Motion of the Machine Problems with the motion of the machine will mainly be caused by: machine setup, alignment, and component failure. • Axis may slow down or move at a feedrate lower than rapid or than the programmed feedrate. • While under motor power, the motion may appear to be "rough" or jerky" . • While under motor power, there may be vibration or noise coming from the X or Z-axis.

#12 lty-source-tcsi-diagnostic-live-tool			故障/报警
部件	live tooling	章节	Problems Relating to Live Tooling Machining Results
起始页	80	安全等级	normal
知识类型	diagnosis	适用机型	TC820LTysi

摘要 | 动力刀具噪声、振动或发热时查驱动对中、轴承、皮带和主轴旋转状态。

A • 标识与来源

记录ID	lty-source-tcsi-diagnostic-live-tool
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725

语言	en
产品系列	TRAK TCsi
B • 型号与位置	
设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	4.6.1
章节	Problems Relating to Live Tooling Machining Results
起始页	80
结束页	80

C • 分类与治理

记录类别	fault
知识类型	diagnosis
部件	live tooling
安全等级	normal
可入库	是
需人工复核	否

D • 业务明细

故障来源	manual_diagnostic_section
故障现象	动力刀具总成噪声/振动过大、过热或动力刀具故障。
原因	动力刀具驱动未对中、驱动轴承损坏或传动皮带故障。
检查项	· 检查动力刀具驱动对中 · 进行振动测试并检查异常振动 · 检查动力刀具主轴是否旋转 · 检查动力刀具驱动皮带
处置措施	检查对中和皮带；疑似轴承或动力刀具故障时联系 TRAK。
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	动力刀具噪声、振动或发热时查驱动对中、轴承、皮带和主轴旋转状态。
关键词	· 动力刀具 · live tooling · 振动 · 过热 · 皮带
检索文本	动力刀具噪声、振动或发热时查驱动对中、轴承、皮带和主轴旋转状态。；动力刀具总成噪声/振动过大、过热或动力刀具故障。；动力刀具驱动未对中、驱动轴承损坏或传动皮带故障。；检查对中和皮带；疑似轴承或动力刀具故障时联系 TRAK。；Live tooling problems may involve excessive noise or vibration, overheating or a faulty live tool. Check live-tool drive alignment, perform a vibration test, inspect the drive belt and verify that the live-tool spindle turns; call TRAK if a bearing or live-tool fault is suspected.; 检查动力刀具驱动对中；进行振动测试并检查异常振动；检查动力刀具主轴是否旋转；检查动力刀具驱动皮带；动力刀具；live tooling；振动；过热；皮带

原文摘录

Possible Cause Check This Tool Probe Set up Tooling problem Make sure the tool probe is set up correctly and calibrated accurately. See Tool Probe section 6.3. Programming Error Improper tooling, loose tooling, tool wear, offsets, Thermal Growth coordinate system, workpiece not properly supported, worn or damaged live center, overhang Gibs too tight or loose from chuck headstock or tailstock. incorrect speeds Inadequate or no Lubrication to Ballscrews and and Feeds. Way surfaces Programmed dimensions are not correct. Check absolute and incremental values. Way surfaces are pocked, scarred, or excessively You must warm the machine up and bring all the worn components to normal operation temperature. See Machine Alignment Warm-Up section 2.17. See Gib Adjustment Chapter 4. Binding or looseness in the drive train or bearings Make sure all the Way surfaces are getting proper lubrication. Check to make sure that the lube pump is functioning properly. Also check for any pinched or blocked oil lines. See Lubrication Chapter 2. Visually check the condition of all the Way surfaces. For machines that may have excessively worn Way surfaces check lubrication to affected areas. After a crash the machine may become out of alignment. See section 4.4. Call TRAK Customer Service at (800)-367-3165 Call TRAK Customer Service at (800)-367-3165 4.6.1 Problems Relating to Live Tooling Machining Results Problems with live tooling machining results can be; machine setup, alignment, and component failure. • Live tooling assembly may have excessive noise or vibrations. • Live tooling assembly is getting too hot. • Live tool may be faulty. Possible Cause Check This The drive is out of alignment. A drive bearing has failed. Check the alignment of the live tooling drive. See section 5.2.2.5. The drive belt is faulty Perform vibration test on the live tool and check for excessive vibration. Call TRAK Customer Service at (800)-367-3165 Check live tooling spindle is turning and inspect the live tooling drive belt. 4.7 Problems Relating to the Machine Tool and Motion of the Machine Problems with the motion of the machine will mainly be caused by: machine setup, alignment, and component failure. • Axis may slow down or move at a feedrate lower than rapid or than the programmed feedrate. • While under motor power, the motion may appear to be "rough" or jerky" . • While under motor power, there may be vibration or noise coming from the X or Z-axis.

#13 lty-source-tcsi-diagnostic-estop			故障/报警
部件	emergency stop circuit	章节	E-Stop Error
起始页	82	安全等级	high
知识类型	diagnosis	适用机型	TC820LTYsi

摘要 | 急停错误先释放所有急停并按住 Reset 1 秒；无效则查急停继电器和主轴箱门开关。

A • 标识与来源

记录ID	lty-source-tcsi-diagnostic-estop
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common

型号专属	否
章节编号	4.9
章节	E-Stop Error
起始页	82
结束页	82

C • 分类与治理

记录类别	fault
知识类型	diagnosis
部件	emergency stop circuit
安全等级	high
可入库	是
需人工复核	否

D • 业务明细

故障来源	manual_diagnostic_section
故障现象	急停错误，机床被禁用。
原因	急停按钮处于按下状态，或急停控制继电器/门开关连接异常。
检查项	· 按住 Reset 1 秒尝试清除 · 确认所有急停按钮均已释放 · 检查电柜内 E-Stop control relay 连接 · 检查 headstock door switch
处置措施	确认无急停被按下后复位；仍不能清除时停止操作并联系 TRAK。
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	急停错误先释放所有急停并按住 Reset 1 秒；无效则查急停继电器和主轴箱门开关。
关键词	· 急停 · E-Stop · Reset · 继电器 · 机床禁用
检索文本	急停错误先释放所有急停并按住 Reset 1 秒；无效则查急停继电器和主轴箱门开关。；急停错误，机床被禁用。；急停按钮处于按下状态，或急停控制继电器/门开关连接异常。；确认无急停被按下后复位；仍不能清除时停止操作并联系 TRAK。；E-stop turns power off to the axis and spindle motors. To clear the message, hold Reset for 1 second. If it does not clear, confirm no E-stops are engaged and check the E-stop control relay connections and headstock door switch; call TRAK if unresolved.; 按住 Reset 1 秒尝试清除；确认所有急停按钮均已释放；检查电柜内 E-Stop control relay 连接；检查 headstock door switch；急停；E-Stop；Reset；继电器；机床禁用

原文摘录

4.9 E-Stop Error (Machine Is Disabled) E stop turns power off to axis and spindle motor when the e stop is pressed a message is displayed on the control as shown in figure 3.9. To clear this message, hold down the reset button for 1 second. If this fail to clear the error, first confirm that no e stops are engaged. In the cabinet check connections to estop control relay. If these are off, then call TRAK Customer Service at (800)-367-3165. Figure 3.9 Emergency stop In the electrical cabinet check connections to E-Stop control relay. If fail to clear issue call TRAK Customer Service at (800)-367-3165. 4.10 Door Guards Not Working Properly Check the switch on top of the door. Check connections to door close relay and door release relay labeled in the electrical cabinet. If the key does not line up with the interlock, the door may have moved out of alignment. Figure 3.10 shows the alarm message when door is not secured. Figure 3.10 Door Error Alarm WARNING!!! If the door does not lock properly, do not operate machine. 4.11 Cable Connections The TCsi machines use several cables to communicate between systems. It is often the case that what appears to be the failure of an electrical component is actually attributable to a poor connection. 1. Turn off and unplug the system from the wall. WARNING! Do not plug and unplug connectors with the system power on. This may cause damage to the connector board and harm to the technician. 2. Visually inspect the connections for excessive debris, moisture, or obvious damage. 3. Carefully clean any chips away from the connectors. 4. One-by-one, take out each connector and then plug them back in. Do the same at the computer/display. 5. Make sure the screws are tightened on each of the connectors. 4.12 Chip Conveyor Troubleshooting The chip conveyor is designed to evacuate chips from the machine envelope after cutting. If not used and maintained correctly issues can occur. When the conveyor is noisy check for a buildup of chips and clear any obstructions. If noise persists check the conveyor belt for damage and call TRAK Customer Service at (800)-367-3165.

#14 lty-source-tcsi-diagnostic-conveyor			故障/报警
部件	chip conveyor	章节	Chip Conveyor Troubleshooting
起始页	82	安全等级	normal
知识类型	diagnosis	适用机型	TC820LTYsi

摘要 | 排屑机噪声先清除切屑和堵塞物，仍异常再检查输送带损坏。

A • 标识与来源

记录ID	lty-source-tcsi-diagnostic-conveyor
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	4.12
章节	Chip Conveyor Troubleshooting

起始页	82
结束页	82

C • 分类与治理

记录类别	fault
知识类型	diagnosis
部件	chip conveyor
安全等级	normal
可入库	是
需人工复核	否

D • 业务明细

故障来源	manual_diagnostic_section
故障现象	排屑机噪声大。
原因	切屑堆积/堵塞或输送带损坏。
检查项	· 清除排屑机内堵塞物 · 检查输送带是否损坏
处置措施	清除切屑后仍有噪声时检查皮带并联系 TRAK。
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	排屑机噪声先清除切屑和堵塞物，仍异常再检查输送带损坏。
关键词	· 排屑机 · chip conveyor · 噪声 · 输送带 · 堵塞
检索文本	排屑机噪声先清除切屑和堵塞物，仍异常再检查输送带损坏。；排屑机噪声大。；切屑堆积/堵塞或输送带损坏。；清除切屑后仍有噪声时检查皮带并联系 TRAK。；When the chip conveyor is noisy, check for a buildup of chips and clear obstructions. If noise persists, check the conveyor belt for damage and call TRAK Customer Service.；清除排屑机内堵塞物；检查输送带是否损坏；排屑机；chip conveyor；噪声；输送带；堵塞

原文摘录

4.9 E-Stop Error (Machine Is Disabled) E stop turns power off to axis and spindle motor when the e stop is pressed a message is displayed on the control as shown in figure 3.9. To clear this message, hold down the reset button for 1 second. If this fail to clear the error, first confirm that no e stops are engaged. In the cabinet check connections to estop control relay. If these are off, then call TRAK Customer Service at (800)-367-3165. Figure 3.9 Emergency stop In the electrical cabinet check connections to E-Stop control relay. If fail to clear issue call TRAK Customer Service at (800)-367-3165. 4.10 Door Guards Not Working Properly Check the switch on top of the door. Check connections to door close relay and door release relay labeled in the electrical cabinet. If the key does not line up with the interlock, the door may have moved out of alignment. Figure 3.10 shows the alarm message when door is not secured. Figure 3.10 Door Error Alarm WARNING!!! If the door does not lock properly, do not operate machine. 4.11 Cable Connections The TCsi machines use several cables to communicate between systems. It is often the case that what appears to be the failure of an electrical component is actually attributable to a poor connection. 1. Turn off and unplug the system from the wall. WARNING! Do not plug and unplug connectors with the system power on. This may cause damage to the connector board and harm to the technician. 2. Visually inspect the connections for excessive debris, moisture, or obvious damage. 3. Carefully clean any chips away from the connectors. 4. One-by-one, take out each connector and then plug them back in. Do the same at the computer/display. 5. Make sure the screws are tightened on each of the connectors. 4.12 Chip Conveyor Troubleshooting The chip conveyor is designed to evacuate chips from the machine envelope after cutting. If not used and maintained correctly issues can occur. When the conveyor is noisy check for a buildup of chips and clear any obstructions. If noise persists check the conveyor belt for damage and call TRAK Customer Service at (800)-367-3165.

#15 lty-source-tcsi-diagnostic-coolant			故障/报警
部件	coolant system	章节	Coolant System
起始页	83	安全等级	normal
知识类型	diagnosis	适用机型	TC820LTYsi

摘要 | 冷却流量低或泵噪声大，检查液位、切屑堵塞、筛网/过滤器和泵叶轮。

A • 标识与来源

记录ID	lty-source-tcsi-diagnostic-coolant
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	4.13
章节	Coolant System

起始页	83
结束页	83

C • 分类与治理

记录类别	fault
知识类型	diagnosis
部件	coolant system
安全等级	normal
可入库	是
需人工复核	否

D • 业务明细

故障来源	manual_diagnostic_section
故障现象	刀塔冷却液流量低，或冷却泵运行噪声大。
原因	系统堵塞、冷却液液位低、切屑堆积、滤网堵塞，或泵电机轴承/叶轮损坏。
检查项	· 检查系统堵塞和冷却液液位 · 检查机床腔体、冷却液箱、筛网和过滤器是否有切屑 · 清洁泵叶轮并转动泵轴
处置措施	清除堵塞并补足冷却液；清洁叶轮和转轴仍无效时可能需要更换泵并联系 TRAK。
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	冷却流量低或泵噪声大，检查液位、切屑堵塞、筛网/过滤器和泵叶轮。
关键词	· 冷却液 · 流量低 · 冷却泵 · 叶轮 · 过滤器
检索文本	冷却流量低或泵噪声大，检查液位、切屑堵塞、筛网/过滤器和泵叶轮。；刀塔冷却液流量低，或冷却泵运行噪声大。；系统堵塞、冷却液液位低、切屑堆积、滤网堵塞，或泵电机轴承/叶轮损坏。；清除堵塞并补足冷却液；清洁叶轮和转轴仍无效时可能需要更换泵并联系 TRAK。；If coolant flow from the turret is low, check for blockages and correct coolant level. If the pump is noisy, check low coolant level or chip buildup in the enclosure, tank, screens and filters. Clear blockages and refill; if cleaning the impeller and rotating the shaft does not fix it, the pump may need replacement.；检查系统堵塞和冷却液液位；检查机床腔体、冷却液箱、筛网和过滤器是否有切屑；清洁泵叶轮并转动泵轴；冷却液；流量低；冷却泵；叶轮；过滤器

原文摘录

4.13 Coolant System The coolant system is distributed from the turret and used to dissipate heat and chips tool and workpiece. If coolant flow is low from the turret check for blockages in the system and ensure the coolant level is at the correct amount. If the pump is noisy while running check for low coolant level or chip build up in the machine enclosure, coolant tank, screens and filters blocking coolant flow. Clear and clean out any blockages and fill up the tank back up. If the pump continues to make noise the motor bearing or impeller could be damage, if cleaning the pump impeller and rotating the shaft does not fix this a new pump may need to be fitted, call TRAK Customer Service at (800)-367-3165. 4.14 Hydraulic System Troubleshooting The hydraulic system drives a number of functions on the machine. When problems occur relating to the hydraulic system the machine will be inoperable. If there is a pressure drop or fluctuation or air bubbles in the hydraulic system check first check the hydraulic fluid level, if the level appears to have dropped check for oil leaks throughout the system. There should enough fluid to prevent air from entering the filter, see section 4.1.3 for the location of the filter in the tank and make sure there is enough fluid to cover the filter. Replace any split hoses, tighten any loose fittings and clean the hydraulic fluid filter before refilling the hydraulic fluid tank. Make sure the hoses are routed correctly and secured in place using zip ties, if hoses rub with other components, have sharp bends, or are kinked, damage can occur. Check the fans and heat exchanger is operating and there are no signs of overheating. If the solenoid system does not operate correctly call TRAK Customer Service at (800)-367-3165.

#16 lty-source-tcsi-diagnostic-hydraulic			故障/报警
部件	hydraulic system	章节	Hydraulic System Troubleshooting
起始页	83	安全等级	high
知识类型	diagnosis	适用机型	TC820LTYsi

摘要 | 液压压力掉压、波动或有气泡时查油位、泄漏、滤芯覆盖、软管和散热。

A • 标识与来源

记录ID	lty-source-tcsi-diagnostic-hydraulic
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	4.14
章节	Hydraulic System Troubleshooting
起始页	83
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C • 分类与治理

记录类别	fault
知识类型	diagnosis
部件	hydraulic system
安全等级	high
可入库	是
需人工复核	否

D • 业务明细

故障来源	manual_diagnostic_section
故障现象	液压压力下降/波动或系统出现气泡，机床无法工作。
原因	液压油位低、漏油、滤芯未被油覆盖、软管破裂/松动/扭结，或散热器过热。
检查项	· 检查液压油位和全系统漏油 · 确保油液足以覆盖过滤器 · 更换裂开的软管、紧固接头并清洁液压滤芯 · 检查风扇和热交换器运行及过热迹象 · 检查软管是否摩擦、急弯或打折
处置措施	补充并更换受损部件，清洁滤芯后重新加注；电磁阀系统仍异常时联系 TRAK。
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	液压压力掉压、波动或有气泡时查油位、泄漏、滤芯覆盖、软管和散热。
关键词	· 液压 · 掉压 · 波动 · 气泡 · 漏油 · 软管
检索文本	液压压力掉压、波动或有气泡时查油位、泄漏、滤芯覆盖、软管和散热。； 液压压力下降/波动或系统出现气泡，机床无法工作。； 液压油位低、漏油、滤芯未被油覆盖、软管破裂/松动/扭结，或散热器过热。； 补充并更换受损部件，清洁滤芯后重新加注；电磁阀系统仍异常时联系 TRAK。； For hydraulic pressure drops, fluctuations or air bubbles, check fluid level and leaks. Ensure enough fluid covers the filter; replace split hoses, tighten loose fittings, clean the filter and refill. Route and secure hoses correctly, check fans and heat exchanger for operation and overheating, and call TRAK if the solenoid system does not operate correctly.; 检查液压油位和全系统漏油；确保油液足以覆盖过滤器；更换裂开的软管、紧固接头并清洁液压滤芯；检查风扇和热交换器运行及过热迹象；检查软管是否摩擦、急弯或打折；液压；掉压；波动；气泡；漏油；软管

原文摘录

4.13 Coolant System The coolant system is distributed from the turret and used to dissipate heat and chips tool and workpiece. If coolant flow is low from the turret check for blockages in the system and ensure the coolant level is at the correct amount. If the pump is noisy while running check for low coolant level or chip build up in the machine enclosure, coolant tank, screens and filters blocking coolant flow. Clear and clean out any blockages and fill up the tank back up. If the pump continues to make noise the motor bearing or impeller could be damage, if cleaning the pump impeller and rotating the shaft does not fix this a new pump may need to be fitted, call TRAK Customer Service at (800)-367-3165.

4.14 Hydraulic System Troubleshooting The hydraulic system drives a number of functions on the machine. When problems occur relating to the hydraulic system the machine will be inoperable. If there is a pressure drop or fluctuation or air bubbles in the hydraulic system check first check the hydraulic fluid level, if the level appears to have dropped check for oil leaks throughout the system. There should enough fluid to prevent air from entering the filter, see section 4.1.3 for the location of the filter in the tank and make sure there is enough fluid to cover the filter. Replace any split hoses, tighten any loose fittings and clean the hydraulic fluid filter before refilling the hydraulic fluid tank. Make sure the hoses are routed correctly and secured in place using zip ties, if hoses rub with other components, have sharp bends, or are kinked, damage can occur. Check the fans and heat exchanger is operating and there are no signs of overheating. If the solenoid system does not operate correctly call TRAK Customer Service at (800)-367-3165.